

Tutorial 3 - Q1

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Question 1 Calculate the autocorrelation function of the ARMA(1,2) process

$$Y_t = 0.6Y_{t-1} + Z_t - 0.3Z_{t-1} - 0.1Z_{t-2}.$$

Using defⁿ 5.1, we know that

AR Coefficients : $\phi_1 = 0.6$

MA Coefficients : $\psi_1 = -0.3$, $\psi_2 = -0.1$

For $\gamma(0)$:

Multiply through by $Y_{t-0} = Y_t$ to get

$$Y_t Y_t = 0.6 Y_{t-1} Y_t + Z_t Y_t - 0.3 Z_{t-1} Y_t - 0.1 Z_{t-2} Y_t$$

Now take the expectation of $Y_t Y_t = Y_t^2$

$$\begin{aligned} \mathbb{E}[Y_t^2] &= 0.6 \mathbb{E}[Y_{t-1} Y_t] + \mathbb{E}[Z_t Y_t] - 0.3 \mathbb{E}[Z_{t-1} Y_t] \\ &\quad - 0.1 \mathbb{E}[Z_{t-2} Y_t] \\ &= \text{Var}[Y_t] + \gamma(\pm 1) - 0.3 \mathbb{E}[Z_{t-1} Y_t] - 0.1 \mathbb{E}[Z_{t-2} Y_t] \\ &= \gamma(0) \end{aligned}$$

$$\Rightarrow \gamma(0) = 0.6 \gamma(\pm 1) + \mathbb{E}[Z_t Y_t] - 0.3 \mathbb{E}[Z_{t-1} Y_t] - 0.1 \mathbb{E}[Z_{t-2} Y_t]$$

For $\gamma(1)$:

Multiply through by Y_{t-1} to get

$$Y_t Y_{t-1} = 0.6 Y_{t-1} Y_{t-1} + Z_t Y_{t-1} - 0.3 Z_{t-1} Y_{t-1} - 0.1 Z_{t-2} Y_{t-1}$$

Take the expectation of $Y_t Y_{t-1}$

$$\begin{aligned} \mathbb{E}[Y_t Y_{t-1}] &= 0.6 \mathbb{E}[Y_{t-1}^2] + \mathbb{E}[Z_t Y_{t-1}] - 0.3 \mathbb{E}[Z_{t-1} Y_{t-1}] - 0.1 \mathbb{E}[Z_{t-2} Y_{t-1}] \\ &= \gamma(\pm 1) + \text{Var}[Y_{t-1}] + \mathbb{E}[Z_t] \mathbb{E}[Y_{t-1}] - 0.3 \mathbb{E}[Z_{t-1} Y_{t-1}] - 0.1 \mathbb{E}[Z_{t-2} Y_{t-1}] \\ &= \gamma(\pm 1) + \gamma(0) + 0 - 0.3 \mathbb{E}[Z_{t-1} Y_{t-1}] - 0.1 \mathbb{E}[Z_{t-2} Y_{t-1}] \end{aligned}$$

$$\gamma(\pm 1) = 0.6 \gamma(0) - 0.3 \mathbb{E}[Z_{t-1} Y_{t-1}] - 0.1 \mathbb{E}[Z_{t-2} Y_{t-1}]$$

$$= \gamma(\pm 1)$$

$$\underline{\gamma(\pm 1) = 0.6 \gamma(0) - 0.3 \mathbb{E}[z_{t-1} y_{t-1}] - 0.1 \mathbb{E}[z_{t-2} y_{t-1}]}$$

For $\gamma(2)$:

Multiply through by y_{t-2} to get:

$$y_t y_{t-2} = 0.6 y_{t-1} y_{t-2} + z_t y_{t-2} - 0.3 z_{t-1} y_{t-1} - 0.1 z_{t-2} y_{t-1}$$

Take the expectation of $y_t y_{t-2}$

$$\mathbb{E}[y_t y_{t-2}] = 0.6 \mathbb{E}[y_{t-1} y_{t-2}] + \underbrace{\mathbb{E}[z_t y_{t-2}]}_{=0} - 0.3 \underbrace{\mathbb{E}[z_{t-1} y_{t-1}]}_{=0} - 0.1 \mathbb{E}[z_{t-2} y_{t-1}]$$

$$\underline{\gamma(\pm 2) = 0.6 \gamma(\pm 1) - 0.1 \mathbb{E}[z_{t-2} y_{t-1}]}$$

For $\gamma(\pm k)$ for $|k| > 2$

$$\begin{aligned} \gamma(\pm k) &= 0.6 \mathbb{E}[y_{t-1} y_{t-k}] \\ &= 0.6 \gamma(\pm(k-1)) \text{ for } |k| > 2 \end{aligned}$$

We now have three unknowns to find,

$$\mathbb{E}[y_t z_t], \mathbb{E}[y_t z_{t-1}], \mathbb{E}[y_t z_{t-2}]$$

$$\begin{aligned} \mathbb{E}[y_t z_t] &= \text{Cov}(y_t, z_t) \\ &= \text{Cov}(z_t, z_t) = \gamma(0) \\ &= \sigma_z^2 \end{aligned}$$

$$\begin{aligned} \mathbb{E}[y_t z_{t-1}] &= \text{Cov}(y_t, z_{t-1}) \\ &= \dots \\ &= 0.6 \mathbb{E}[y_{t-1} z_{t-1}] - 0.3 \mathbb{E}[z_{t-1}^2] \\ &= 0.6 \gamma(0) - 0.3 \gamma(0) \\ &= 0.3 \sigma_z^2 \end{aligned}$$

$$\begin{aligned} \mathbb{E}[y_t z_{t-2}] &= \text{Cov}(y_t, z_{t-2}) \\ &= \dots \\ &= 0.6 \mathbb{E}[y_{t-1} z_{t-2}] - 0.1 \mathbb{E}[z_{t-2}^2] \\ &= (0.6)(0.3 \sigma_z^2) - 0.1(\sigma_z^2) \\ &= 0.08 \sigma_z^2 \end{aligned}$$

$$\begin{aligned}
 &= (0.6)(0.3\sigma_z^2) - 0.1(\sigma_z^2) \\
 &= 0.08\sigma_z^2
 \end{aligned}$$

$\therefore$ we have the following equations

$$\gamma(0) = 0.6\gamma(\pm 1) + \sigma_z^2 - 0.3(0.3\sigma_z^2) - 0.1(0.08\sigma_z^2)$$

$$\gamma(0) = 0.6\gamma(\pm 1) + 0.902\sigma_z^2$$

$$\gamma(\pm 1) = 0.6\gamma(0) - 0.3(\sigma_z^2) - 0.1(0.3\sigma_z^2)$$

$$\gamma(\pm 1) = 0.6\gamma(0) - 0.33\sigma_z^2$$

Now let's solve them,

$$\gamma(0) = 0.6[0.6\gamma(0) - 0.33\sigma_z^2] + 0.902\sigma_z^2$$

$$\Rightarrow \gamma(0) = 0.36\gamma(0) + 0.704\sigma_z^2$$

$$\Rightarrow \gamma(0)(1 - 0.36) = 0.704\sigma_z^2$$

$$\Rightarrow \gamma(0) = 1.1\sigma_z^2$$

Subbing $\gamma(0)$ into $\gamma(\pm 1)$ we get,

$$\begin{aligned}
 \gamma(\pm 1) &= 0.6(1.1\sigma_z^2) - 0.33\sigma_z^2 \\
 &= 0.33\sigma_z^2
 \end{aligned}$$

$$\begin{aligned}
 \gamma(\pm 2) &= (0.6)(0.33\sigma_z^2) - (0.1)\sigma_z^2 \\
 &= 0.098\sigma_z^2
 \end{aligned}$$

$$\gamma(\pm k) = 0.6\gamma(\pm k-1) \text{ for } |k| > 2$$

$\therefore$ we have the following ACF

$$\rho(\pm k) = \begin{cases} 1 & , \text{ for } k = 0 \\ 0.3 & , \text{ for } |k| = 1 \\ 0.084 & , \text{ for } |k| = 2 \\ (0.6)^{k-2} \rho(\pm 2) & , \text{ for } |k| > 2 \end{cases}$$